

LSC Lab 6 – Kubernetes

Report

Szymon Ożóg 412040

Konfiguracja AWS CLI

aws configure

Instalacja helm-a:

choco install kubernetes-helm

Utworzenie klastra – przez aws academy

Utworzenie grupy węzłów

aws eks create-nodegroup --cluster-name lsc-cluster --nodegroup-name lsc-ng --node-role arn:aws:iam::363526937044:role/LabRole --subnet subnet-0ac2fdb7d16be0d0d subnet-053c86760f165cd88 --instance-types t3.medium --scaling-config minSize=1,maxSize=2,desiredSize=1 --disk-size 20 --ami-type AL2_x86_64

```
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>aws eks create-nodegroup --cluster-name lsc-cluster --nodegroup-name lsc-ng --node-role arn:aws:iam::363526937044:role/LabRole --subnet subnet-0ac2fdb7d16be0d0d subnet-053c86760f165cd88 --instance-types t3.medium --scaling-config minSize=1,maxSize=2,desiredSize=1 --disk-size 20 --ami-type AL2_x86_64
{
  "nodegroup": {
    "nodegroupName": "lsc-ng",
    "nodegroupArn": "arn:aws:eks:us-east-1:363526937044:nodegroup/lsc-cluster/lsc-ng/52cb1a53-be7c-f6ae-cb6d-5842027fd062",
    "clusterName": "lsc-cluster",
    "version": "1.32",
    "releaseVersion": "1.32.1-20250403",
    "createdAt": "2025-04-14T16:15:14.738000+02:00",
    "modifiedAt": "2025-04-14T16:15:14.738000+02:00",
    "status": "CREATING",
    "capacityType": "ON_DEMAND",
    "scalingConfig": {
      "minSize": 1,
      "maxSize": 2,
      "desiredSize": 1
    },
    "instanceTypes": [
      "t3.medium"
    ],
    "subnets": [
      "subnet-0ac2fdb7d16be0d0d",
      "subnet-053c86760f165cd88"
    ],
    "amiType": "AL2_x86_64",
    "nodeRole": "arn:aws:iam::363526937044:role/LabRole",
    "diskSize": 20,
    "health": {
      "issues": []
    },
    "updateConfig": {
      "maxUnavailable": 1
    },
    "tags": {}
  }
}
```

Konfiguracja kubectl

```
aws eks --region us-east-1 update-kubeconfig --name lsc-cluster
```

```
kubectl get nodes
```

```
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>aws eks --region us-east-1 update-kubeconfig --name lsc-cluster
Added new context arn:aws:eks:us-east-1:363526937044:cluster/lsc-cluster to C:\Users\Szymon\.kube\config

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-172-31-35-98.ec2.internal        Ready    <none>   34s   v1.32.1-eks-5d632ec

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>
```

Dodanie repozytorium Helm

```
helm repo add nfs-ganesha-server-and-external-provisioner https://kubernetes-sigs.github.io/nfs-ganesha-server-and-external-provisioner/
```

```
helm repo update
```

```
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>helm repo add nfs-ganesha-server-and-external-provisioner https://kubernetes-sigs.github.io/nfs-ganesha-server-and-external-provisioner/
"nfs-ganesha-server-and-external-provisioner" has been added to your repositories

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "nfs-ganesha-server-and-external-provisioner" chart repository
Update Complete. 🎉Happy Helming!🎉
```

Instalacja NFS Provisioner

```
helm install nfs-server-provisioner nfs-ganesha-server-and-external-provisioner/nfs-server-provisioner -f nfs-values.yaml
```

```
kubectl get storageclass
```

```
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>helm install nfs-server-provisioner nfs-ganesha-server-and-external-provisioner/nfs-server-provisioner -f nfs-values.yaml
NAME: nfs-server-provisioner
LAST DEPLOYED: Mon Apr 14 16:21:10 2025
NAMESPACE: default
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
The NFS Provisioner service has now been installed.

A storage class named 'nfs' has now been created
and is available to provision dynamic volumes.

You can use this storageclass by creating a `PersistentVolumeClaim` with the
correct storageClassName attribute. For example:

---
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  name: test-dynamic-volume-claim
spec:
  storageClassName: "nfs"
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 100Mi

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl get storageclass
NAME      PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION   AGE
gp2       kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer false                  16m
nfs       cluster.local/nfs-server-provisioner Delete          Immediate            true                   33s
```

Kubernetes setup repository: <https://github.com/SzymixOz/lsc-kubernetes/>

Setup files application:

kubectl apply -f pvc.yaml

kubectl get pvc

```
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl apply -f pvc.yaml
persistentvolumeclaim/nfs-pvc created

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl get pvc
NAME          STATUS    VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS
AGE
nfs-pvc       Bound     pvc-7bf2e74a-47df-4c90-a3b0-944c68521bde  1Gi        RWX            nfs            <unset>
23s

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>
```

kubectl apply -f deployment.yaml

kubectl get pods

```
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nfs-server-provisioner-0            1/1     Running   0           31m
nginx-deployment-777fffbb79-2hwr6  1/1     Running   0           14s
```

kubectl apply -f service.yaml

kubectl get service

```
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl apply -f service.yaml
service/nginx-service created

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl get service
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes    ClusterIP   10.100.0.1    <none>         <none>            48m
nfs-server-provisioner  ClusterIP   10.100.55.29  <none>         2049/TCP,2049/UDP,32803/TCP,32803/UDP,20048/TCP,20048/UDP,875/TCP,875/UDP,111/TCP,111/UDP,662/TCP,662/UDP  32m
nginx-service  LoadBalancer  10.100.167.174  a8447aabc6f6f4e4fab4448df12117f1-2020613130.us-east-1.elb.amazonaws.com  80:31927/TCP  15s

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>
```

kubectl apply -f job.yaml

kubectl get jobs

kubectl logs job/copy-content-job

```
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl apply -f job.yaml
job.batch/copy-content-job created

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl get jobs
NAME             STATUS    COMPLETIONS   DURATION   AGE
copy-content-job Complete   1/1           3s         12s

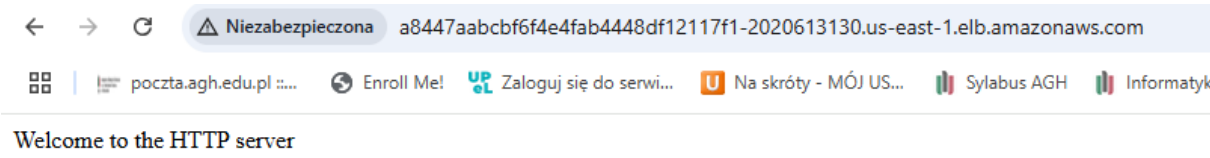
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl logs job/copy-content-job

D:\Studia\Magisterskie\Semestr_1\LSC\lab6>
```

Testowanie serwera HTTP

kubectl get service nginx-service

```
D:\Studia\Magisterskie\Semestr_1\LSC\lab6>kubectl get service nginx-service
NAME      TYPE        CLUSTER-IP      EXTERNAL-IP      PORT(S)
nginx-service  LoadBalancer  10.100.167.174   a8447aabcbf6f4e4fab4448df12117f1-2020613130.us-east-1.elb.amazonaws.com  80:31927/TCP  4m57s
```



Screenshot from final cluster workloads:

Workloads: Pods (12)

View details

Pod is the smallest and simplest Kubernetes object. A Pod represents a set of running containers on your cluster. [Learn more](#)

All Namespaces

Filter Pods by name

Name	Created
aws-node-rprs	an hour ago
copy-content-job-92qht	7 minutes ago
coredns-6b9575c64c-2kvgw	an hour ago
coredns-6b9575c64c-zzfd4	an hour ago
eks-node-monitoring-agent-n7wsg	an hour ago
eks-pod-identity-agent-djfhm	an hour ago
external-dns-6b6d548589-4mk8g	an hour ago
kube-proxy-hrk9n	an hour ago
metrics-server-57b774cc8d-4p8qj	an hour ago
metrics-server-57b774cc8d-qxf8z	an hour ago

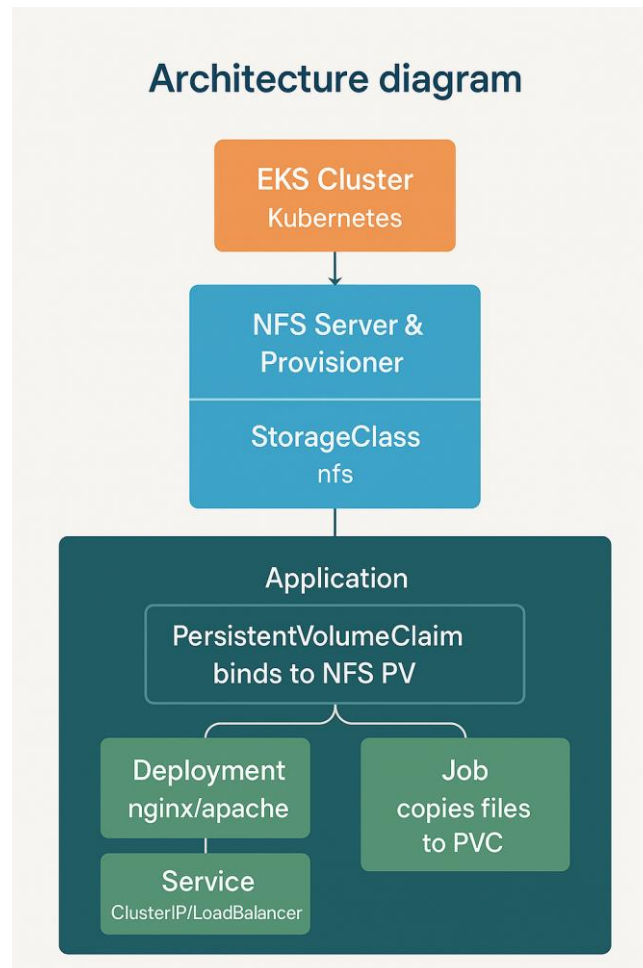


Diagram explanation:

- **EKS Cluster:** The foundational Kubernetes setup hosted on AWS using Elastic Kubernetes Service.
- **NFS Server & Provisioner:** Supplies a shared filesystem for Kubernetes pods through the NFS protocol. The provisioner ensures that PersistentVolumes (PVs) are created automatically based on demand.
- **StorageClass (nfs):** A template defining how dynamic volumes should be provisioned using the NFS backend.
- **PersistentVolumeClaim (PVC):** A resource used by applications to request specific storage space from the dynamically created NFS volumes.
- **Deployment:** Hosts a web server (such as nginx or Apache) with the NFS volume mounted to serve static content.
- **Job:** A one-time task that writes a static HTML file (like index.html) into the shared NFS-mounted volume for the web server to serve.
- **Service:** Makes the deployed web server accessible, either internally within the cluster or externally via a LoadBalancer.